

Muhammad Haris Imran

AI Backend Engineer

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PROFESSIONAL SUMMARY

AI Software Engineer with production experience building LLM-powered enterprise applications, RAG pipelines, multi-agent orchestration, multimodal image generation, and real-time streaming infrastructure. Currently engineering at INTECH Automation & Intelligence, building backend AI systems for enterprise clients across GPT-4, Claude, and Gemini. Previously built data infrastructure with Snowflake and SQL, and developed a deep learning diagnostic system achieving 93% accuracy.

PROFESSIONAL EXPERIENCE

AI Software Engineer

INTECH Automation & Intelligence, Lahore | Dec 2025 - Present

- Engineered production RAG pipelines with document grounding, language detection, and strict PDF-to-answer traceability for enterprise knowledge management.
- Built LangGraph ReAct agent workflows with multi-step tool orchestration, LLM interrupt handling, and stateful checkpointing for complex enterprise tasks.
- Integrated multi-model LLM support across GPT-4, Claude, and Gemini with unified callback handlers for consistent token tracking across all providers.
- Implemented real-time token-level streaming over Azure Web PubSub using custom LangChain callbacks, delivering progressive answers to the frontend.
- Built image generation and editing tools for LangGraph agents using GPT Image 2.5 and Google Nano Banana, with multi-turn edits on generated images, safety-block handling, and secure delivery via Azure Blob Storage SAS URLs.
- Developed a production-grade web fetch tool with SSRF protection, redirect safety validation, and HTML sanitization for secure LLM agent browsing.
- Debugged and resolved production issues including retry logic, WebSocket lifecycle management, and environment-specific RAG behavior mismatches.

Data Analyst

Logi Vertex | Jan 2024 - Sep 2024

- Executed complex SQL queries in Snowflake to streamline ETL workflows, build semantic data models, and optimize data pipeline performance.
- Conducted exploratory data analysis to surface key business patterns and deliver actionable recommendations to stakeholders.
- Collaborated cross-functionally to translate business requirements into technical data solutions, improving reporting efficiency.

Data Analyst / WordPress Developer

Web Gait | Jan 2023 - Dec 2023

- Performed data cleaning, transformation, and validation to support accurate business reporting and analytics workflows.
- Built and maintained WordPress websites with responsive design and SEO optimization, improving site performance and user engagement.

OPEN SOURCE PROJECTS

DocChat — RAG Chatbot with Evaluation Pipeline

github.com/harischughtai/docchat | *LangChain, FAISS, OpenAI, Gradio, Python*

- Production-grade RAG chatbot with PDF ingestion, MMR-based retrieval, streaming responses, and built-in evaluation metrics (Faithfulness, Relevancy, Context Precision) using LLM-as-judge.

AgentFlow — Multi-Agent Orchestrator

github.com/harischughtai/agentflow | *LangGraph, LangChain, Tavily, Gradio, Python*

- Multi-agent system using LangGraph where a Supervisor Agent coordinates Research and Writer agents with stateful checkpointing, decision logging, error recovery, and human-in-the-loop approval.

ModelRouter — Multi-Model LLM Gateway

github.com/harischughtai/modelrouter | *FastAPI, LangChain, Tiktoken, Streamlit, Python*

- Intelligent LLM gateway routing queries to optimal model (GPT-4o, Claude, Gemini) based on complexity scoring and cost sensitivity, with automatic fallback and real-time cost dashboard.

PixelRouter — Multi-Model Image Generation

github.com/harischughtai/pixelrouter | *OpenAI GPT Image, Gemini (Nano Banana), LangChain, Gradio, Python*

- Unified interface for GPT Image and Nano Banana with a provider-agnostic design, concurrent side-by-side model comparison, multi-turn editing by image ID exposed as LangChain tools, and token and latency tracking.

SafeFetch — Secure Web Fetcher for LLM Agents

github.com/harischughtai/safefetch | *Python, aiohttp, LangChain, Security*

- Production-grade web fetcher with SSRF protection, post-redirect re-validation, streaming size cap, HTML sanitization, and prompt injection boundary wrapping. Includes LangChain tool wrapper.

ROP Detection System — Medical Imaging AI

Deployed on Hugging Face Spaces | *PyTorch, TensorFlow, VGG16, SegNet, OpenCV, Gradio*

- Deep learning diagnostic system for detecting Retinopathy of Prematurity in retinal images using hybrid VGG16 + SegNet architecture, achieving 93% classification accuracy.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, SQL, Dart, GDScript, Bash
AI & LLM:	LangChain, LangGraph, RAG, OpenAI, Anthropic Claude, Gemini, Agents & MCP, Embeddings, Vector Search, Fine-tuning, Prompt Engineering, Multimodal AI, Image Generation (GPT Image, Nano Banana)
ML & CV:	PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, MediaPipe, Hugging Face
Backend:	FastAPI, Node.js, MongoDB, PostgreSQL, Supabase, Vector DBs, REST APIs, JWT, NextAuth
Frontend:	React, Next.js, Flutter, Tailwind CSS, MUI, Streamlit, Gradio
DevOps:	Docker, Azure (Blob Storage, Web PubSub), AWS, Google Cloud, Vercel, Render, GitHub Actions, CI/CD, Linux, Git/Git LFS
Testing:	Playwright, Selenium
Data:	Snowflake, MySQL, SQL Server, ETL, Power BI, Tableau, Pandas, NumPy

EDUCATION

Bachelor of Science in Computer Science

University of Central Punjab (UCP), Lahore | Graduated June 2025